ENR 3. ATS ROUTES

ENR 3.1 CONVENTIONAL NAVIGATION ROUTES

3.1.1 INTERNATIONAL ATS ROUTES

Route designator (RNP/RNAV) Name of significant points Coordinates	Track MAG↑/↓ VOR RDL DIST (COP)		Upper limits / Lower limits or Minimum altitude(MOCA) ft AMSL or FL Airspace classification	Lateral limits NM	Direction of cruising levels Odd	Direction of cruising levels Even	Remarks Controlling unit channel Logon address
1	2		3	4	5		6
A582							
△ ANYANG VORTAC(SEL) 372449N 1265542E							**Daegu ACC**
	174°/354° 12.4		UNL / 4 500 (3 400) Class A, D, G	10	↓		FREQ : (at or below FL 295) 128.70(118.925) MHz 270.50(263.60) MHz FREQ : (above FL 295) 132.80(120.525) MHz 290.60(335.45) MHz
△ POLEG 371249N 1265935E	174°/354° 7.4						**Airspace Classification** Class A : Above FL 200 - FL 600 Class D : MEA - FL 200 Class G : Above FL 600 - UNL
△ SONGTAN VORTAC(SOT) 370540N 1270154E	144°/324° 21.6		UNL / 8 000 (3 300) Class A, D, G				
▲ OSPOT 365018N 1272055E	144°/324° 10.4						**Daegu ACC**
▲ VASLI 364252N 1273003E	144°/324° 21.9	(69/39)	UNL / 8 000 (4 300) Class A, D, G				FREQ : 125.375(125.775, 124.575) MHz 234.15(317.35, 335.50) MHz
▲ MAKDU 362712N 1274909E	144°/324° 14.6		UNL / 8 000 (4 400) Class A, D, G				**Airspace Classification** Class A : Above FL 200 - FL 600 Class D : MEA - FL 200 Class G : Above FL 600 - UNL
▲ BITUX 361645N 1280148E	144°/325° 39.2		UNL / 10 000 (5 200) Class A, D, G				
△ DALSEONG VORTAC(TGU) 354835N 1283527E	162°/342° 20.4		UNL / 6 000 (5 400) Class A, D, G				
▲ KALOD 353012N 1284626E	163°/343° 25.4						
△ BUSAN VORTAC(PSN) 350721N 1285958E	163°/343° 26.6		UNL / 4 000 (3 000) Class A, D, G			↑	
▲ APELA(FIR BDRY) 344323N 1291400E							
INCHEON FIR							
FUKUOKA FIR							

Change : Information of track MAG for A582.

Route designator (RNP/RNAV) Name of significant points Coordinates	Track MAG ↑ / ↓ VOR RDL DIST (COP)	Upper limits / Lower limits or Minimum altitude(MOCA) ft AMSL or FL Airspace classification	Lateral limits NM	Direction of cruising levels Odd	Direction of cruising levels Even	Remarks Controlling unit channel Logon address
1	2	3	4	5		6
A586*						
△ TENAS 373820N 1313427E	228°/047° 68.1	UNL / FL 310(4 600) Class A, G	10		↓	**Daegu ACC** FREQ : 122.250(125.925) MHz 263.350(263.85) MHz Airspace Classification refer to ENR 3.1-1
△ AGSUS 364521N 1304044E	229°/049° 36.2	UNL / FL 310(1 500) Class A, G				
▲ DABIK 361743N 1301143E	228°/048° 28.1					**Daegu ACC** FREQ : 120.575(119.375, 119.325, 134.375) MHz 254.70(335.75) MHz Airspace Classification refer to ENR 3.1-1
▲ BULGA 355609N 1294924E	228°/048° 40.2	UNL / FL 310(2 900) Class A, G				
▲ BEDOM 352513N 1291754E	228°/048° 23.1	UNL / FL 310(4 000) Class A, G				
△ BUSAN VORTAC(PSN) 350721N 1285958E	238°/058° 10.4	(79/78) / UNL** / 11 000 / 10 000 / 8 000(3 500) Class D				**Daegu ACC** FREQ : 125.375(125.775, 124.575) MHz 234.15(317.35, 335.50) MHz Airspace Classification refer to ENR 3.1-1
△ OMOTU 350033N 1285022E	238°/057° 22.3	10 000 / 8 000(3 200) Class D				
▲ TOPAX 344555N 1282952E	237°/057° 46.4	10 000 / 8 000(2 800) Class D				**Daegu ACC** FREQ : 128.175(128.325) MHz 335.50(275.20) MHz Airspace Classification refer to ENR 3.1-1
▲ GOSBO 341517N 1274734E	237°/056° 30.7	10 000 / 8 000(2 000) Class D				
▲ MAKET 335452N 1271953E	236°/056° 17.3	10 000 / 8 000(1 500) Class D				**Incheon ACC** FREQ : 124.525(132.425) MHz 255.40(233.50, 348.10) MHz Airspace Classification refer to ENR 3.1-1
△ ATINA 334320N 1270423E	236°/056° 10.3	10 000 / 8 000(1 500) Class D				
△ MANOL 333629N 1265514E	236°/056° 20.0	10 000 / 9 000(8 700) Class D				
△ JEJU VORTAC(CJU) 332305N 1263727E	170°/350° 24.0	UNL / 9 000(8 700) Class A, D, G				※ The cruising levels from CJU to RUGMA are even levels due to operational reasons.
▲ TOSAN 330012N 1264619E	170°/350° 31.5	UNL / 9 000(1 500) Class A, D, G		↑		※ The cruising levels from RUGMA to CJU are odd levels due to operational reasons.
▲ RUGMA(FIR BDRY) 323012N 1265753E						
INCHEON FIR						
FUKUOKA FIR						

TENAS-PSN CDR1 Operational hour(UTC) - Weekdays : 1100-2200 - SAT: 2200 on the preceding until 2400 on the Saturday - SUN : 0000-2200 - Holiday : 1100 on the preceding until 2200 on the holiday. Rest of A586-PERM. See ENR 1.1-1.2.

* A586(PSN-CJU) is only used for Non-RNAV aircraft. Any aircraft approved for RNAV operations should use Y571 or Y572.
** Any aircraft planning to operate above 10 000 ft between PSN and CJU must coordinate with Daegu ACC before flight planning.

Change : Information of track MAG for A586.

Route designator (RNP/RNAV) Name of significant points Coordinates	Track MAG↑/↓ VOR RDL DIST (COP)	Upper limits / Lower limits or Minimum altitude(MOCA) ft AMSL or FL / Airspace classification	Lateral limits NM	Direction of cruising levels Odd	Direction of cruising levels Even	Remarks Controlling unit channel Logon address
1	2	3	4	5		6
FUKUOKA FIR						
INCHEON FIR						
A593						
▲ ONIKU(FIR BDRY) 321142N 1263917E						**Incheon ACC**
	263° / 082° 31.4	UNL / FL 240 (1 500) Class A, G	50		↓	FREQ : (at or above FL 335) 133.425(132.425) MHz 234.35(234.65) MHz
△ NIRAT 320354N 1260329E	263° / 083° 14.5					FREQ : (below FL 335) 125.725(132.825, 128.375) MHz 232.95(233.15) MHz
△ PONIK 320021N 1254659E	262° / 082° 41.4					Airspace Classification refer to ENR 3.1-1
▲ SADLI 314948N 1250000E	263° / 082° 52.8			↑	↓	※ Eastbound from SADLI to ONIKU is only for Non-approval of RNAV 2. Any aircraft approved for RNAV 2 operation should use Y590.
▲ LAMEN(FIR BDRY) 313636N 1240000E						
INCHEON FIR						
SHANGHAI FIR						

1. Delegation
The responsibility of Air Traffic Services in the scope below within Incheon FIR is delegated to Shanghai ACC.
- Horizontal scope : 320229N 1240000E - 321540N 1250000E - 312356N 1250000E - 311043N 1240000E
- Vertical scope : FL 240 through FL 410 inclusive

2. Flight Level Allocation Scheme (FLAS)
(1) The west of SADLI
- Eastbound : FL 250, FL 270, FL 290, FL 310, FL 330, FL 350, FL 370, FL 390, FL 410
- Westbound : FL 240, FL 260, FL 280, FL 300, FL 320, FL 340, FL 360, FL 380, FL 400

(2) The east of SADLI
- Eastbound(From Shanghai FIR To Fukuoka FIR) : FL 250, FL 290, FL 310, FL 390
- Westbound(From Fukuoka FIR To Shanghai FIR) : FL 240, FL 280, FL 300, FL 400

A595						
△ JEJU VORTAC(CJU) 332305N 1263727E						**Incheon ACC**
	089° / 270° 35.9	UNL / 9 000 (8 700) Class A, D, G	8	↓		FREQ : 124.525(132.425) MHz 255.40(233.50, 348.10) MHz
▲ TAMNA 332815N 1271953E	090° / 270° 49.8	UNL / 9 000 (1 500) Class A, D, G			↑	Airspace Classification refer to ENR 3.1-1
▲ SAMDO(FIR BDRY) 333503N 1281857E						
INCHEON FIR						
FUKUOKA FIR						

Change : Information of track MAG for A595.

Route designator (RNP/RNAV) Name of significant points Coordinates	Track MAG↑/↓ VOR RDL DIST (COP)	Upper limits / Lower limits or Minimum altitude(MOCA) ft AMSL or FL Airspace classification	Lateral limits NM	Direction of cruising levels Odd	Direction of cruising levels Even	Remarks Controlling unit channel Logon address
1	2	3	4	5		6
PYONGYANG FIR						
INCHEON FIR						
B332						
▲ KANSU(FIR BDRY) 383800N 1322830E	180°/360° 40.2	UNL / FL 200 (1 500) Class A, G	50	↓		**Daegu ACC** FREQ : 122.250(125.925) MHz 263.350(263.85) MHz Airspace Classification refer to ENR 3.1-1
△ PALDU 375813N 1323625E	180°/360° 19.9					
△ SABET 373829N 1324019E	180°/360° 19.9				↑	
▲ IGRAS(FIR BDRY) 371846N 1324411E						
INCHEON FIR						
FUKUOKA FIR						
B467						
△ GANGWON VORTAC(KAE) 374203N 1284514E	100°/280° 44.3	UNL / 8 000 (7 100) Class A, D, G	10	↓		**Daegu ACC** FREQ: (at or below FL 295) 134.175(123.65) MHz 272.40(233.60) MHz FREQ : (above FL 295) 122.250(125.925) MHz 263.350(263.85) MHz Airspace Classification refer to ENR 3.1-1
△ LESBU 374116N 1294104E	100°/281° 7.8	UNL / 8 000 (1 500) Class A, D, G				
▲ UGOVI 374105N 1295051E	101°/281° 20.1	UNL / 8 000 (1 500) Class A, D, G				**Daegu ACC** FREQ : 122.250(125.925) MHz 263.350(263.85) MHz Airspace Classification refer to ENR 3.1-1
△ BUSKO 374033N 1301610E	101°/282° 62.2	UNL / 8 000 (4 600) Class A, D, G				
△ TENAS 373820N 1313427E	044°/224° 20.0	UNL / FL 200 (1 500) Class A, G	50			
△ MALSO 375440N 1314904E	044°/225° 53.3				↑	
▲ KANSU(FIR BDRY) 383800N 1322830E						
INCHEON FIR						
PYONGYANG FIR						

Change : Information of track MAG for B467.

Route designator (RNP/RNAV) Name of significant points Coordinates	Track MAG↑/↓ VOR RDL DIST (COP)		Upper limits / Lower limits or Minimum altitude(MOCA) ft AMSL or FL / Airspace classification		Lateral limits NM	Direction of cruising levels Odd	Direction of cruising levels Even	Remarks Controlling unit channel Logon address
1	2		3		4	5		6
B576*								**Daegu ACC**
△ ANYANG VORTAC(SEL) 372449N 1265542E	174°/354° 12.4		UNL / 4 500 (3 400) Class A, D, G		10		↓	FREQ : (at or below FL 295) 128.70(118.925) MHz 270.50(263.60) MHz FREQ : (above FL 295) 132.80(120.525) MHz 290.60(335.45) MHz Airspace Classification refer to ENR 3.1-1
△ POLEG 371249N 1265935E	174°/354° 7.4							
△ SONGTAN VORTAC(SOT) 370540N 1270154E	194°/014° 21.5	(59/59) KUZ R-060/ D 19.1	13 000 / 7 000 Class D	(2 800)				**Incheon ACC** FREQ : (at or below FL 255) 126.175(134.375) MHz 317.85(335.55) MHz FREQ : (above FL 255) 132.15(123.55) MHz 263.15(272.60) MHz Airspace Classification refer to ENR 3.1-1 * Note B576 is only used for Non-RNAV aircraft. Any aircraft approved for RNAV operations should use Y711 or Y722.
▲ OLMEN 364413N 1265928E	194°/014° 21.1			(3 600)				
▲ ENTEL 362311N 1265705E	194°/013° 29.4			(3 100)				
△ RINBO 355352N 1265349E	193°/013° 22.7		UNL** / FL 140 Class A, D, G	(3 400)				
▲ LINTA 353116N 1265119E	193°/013° 23.8			(4 700)				**Incheon ACC** FREQ : (at or below FL 255) 120.725(128.30) MHz 263.90(272.75) MHz FREQ : (above FL 255) 123.725(124.50) MHz 239.25(275.40) MHz Airspace Classification refer to ENR 3.1-1
△ GWANGJU VOR(KWA) 350734N 1264844E	193°/013° 52.5	(52/52)	13 000 / 8 000 (4 700) Class D	UNL** / FL 140				
▲ IPDAS 341515N 1264301E	193°/013° 52.3		13 000 / 9 000 (8 700) Class D					**Incheon ACC** FREQ : 124.525(132.425) MHz 255.40(233.50, 348.10) MHz Airspace Classification refer to ENR 3.1-1 * Note Any aircraft planning to operate above FL 140 between SOT and CJU must coordinate with Incheon ACC before flight planning. ** Note Any southbound aircraft planning to operate below FL 320 between CJU and ATOTI must use Y711 until BONSO, then use Z86 to ATOTI.
△ JEJU VORTAC(CJU) 332305N 1263727E	208°/028° 24.3		UNL / 9 000 (8 700) Class A, D, G					
▲ SOSDO 330012N 1262735E	208°/028° 29.5		UNL / 8 000 (1 500) Class A, D, G					
△ SAMLO 323223N 1261536E	208°/028° 30.2							**Incheon ACC** FREQ : (at or above FL 335) 133.425(132.425) MHz 234.35(234.65) MHz FREQ : (below FL 335) 125.725(132.825, 128.375) MHz 232.95(233.15) MHz Airspace Classification refer to ENR 3.1-1
△ NIRAT 320354N 1260329E	208°/028° 18.1							
△ ELGEP 314653N 1255617E	208°/027° 12.1							
▲ TESIM 313526N 1255128E	208°/027° 100.9							
▲ ATOTI(FIR BDRY) 300013N 1251154E						↑		
INCHEON FIR								
FUKUOKA FIR								

NAVAID(DME) GAP : Between 44 NM from KWA and 55 NM from SOT, below 10 000 ft AMSL, request to controller is required to get distance information.

Change : Information of track MAG for B576.

Route designator (RNP/RNAV) Name of significant points Coordinates	Track MAG↑/↓ VOR RDL DIST (COP)	Upper limits / Lower limits or Minimum altitude(MOCA) ft AMSL or FL / Airspace classification	Lateral limits NM	Direction of cruising levels: Odd	Direction of cruising levels: Even	Remarks Controlling unit channel Logon address
1	2	3	4	5		6
G339 △ BUSAN VORTAC(PSN) 350721N 1285958E						**Daegu ACC** FREQ : 125.375(125.775, 124.575) MHz 234.15(317.35, 335.50) MHz Airspace Classification refer to ENR 3.1-1
	150°/330° 25.6	UNL / 10 000 (3 400) Class A, D, G	8	↓	↑	
▲ INVOK(FIR BDRY) 344719N 1291923E INCHEON FIR						
FUKUOKA FIR						

Change : Information of track MAG for G339.

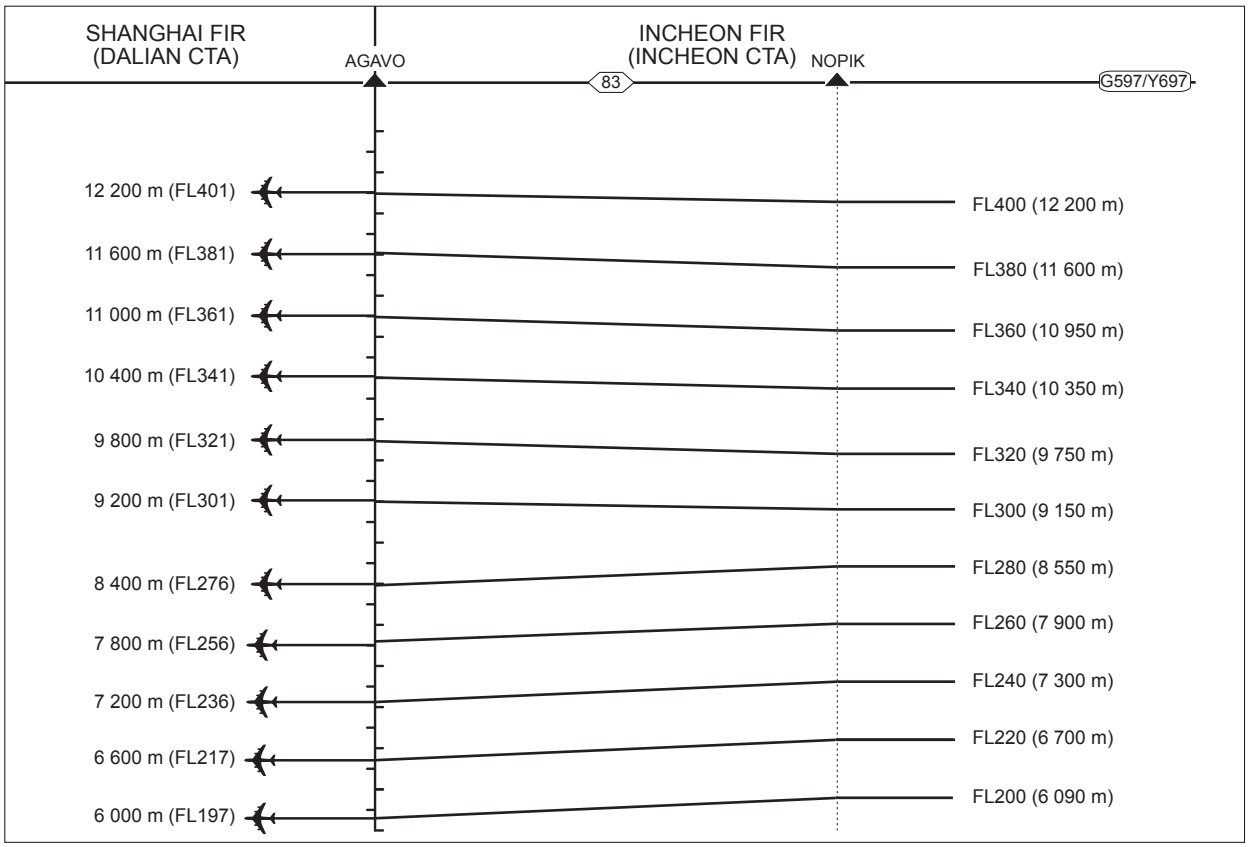
Route designator (RNP/RNAV) Name of significant points Coordinates	Track MAG↑/↓ VOR RDL DIST (COP)		Upper limits / Lower limits or Minimum altitude(MOCA) ft AMSL or FL Airspace classification	Lateral limits NM	Direction of cruising levels Odd	Direction of cruising levels Even	Remarks Controlling unit channel Logon address
1	2		3	4	5		6
G585							**Daegu ACC**
△ ANYANG VORTAC(SEL) 372449N 1265542E				8	↓		FREQ : (at or below FL 295) 128.70(118.925) MHz 270.50(263.60) MHz FREQ : (above FL 295) 132.80(120.525) MHz 290.60(335.45) MHz Only flying westbound from KPO to SEL on G585 is authorized except ACFT departing from RKTY or RKTI. Aircraft flying eastbound from SEL to KPO at or above 11 000 ft on G585 shall get PPR 24 hours before from Daegu ACC. No PPR is needed at or below 10 000 ft. Airspace Classification refer to ENR 3.1-1
	133°/313° 10.7		UNL / 8 000 (3 400) Class A, D, G				
△ KALMA 371845N 1270645E							
	133°/314° 19.3						
△ KAKSO 370745N 1272637E							
	134°/314° 11.5	(50/32)	UNL / 8 000 (3 600) Class A, D, G				
▲ GUKDO 370111N 1273823E							**Daegu ACC** FREQ : 125.375(125.775, 124.575) MHz 234.15(317.35, 335.50) MHz Only flying westbound from KPO to SEL on G585 is authorized except ACFT departing from RKTY or RKTI. Aircraft flying eastbound from SEL to KPO at or above 11 000 ft on G585 shall get PPR 24 hours before from Incheon/Daegu ACC. No PPR is needed at or below 10 000 ft. Airspace Classification refer to ENR 3.1-1
	134°/314° 9.2		UNL / 8 000 (3 700) Class A, D, G				
△ ENSAL 365554N 1274747E							
	134°/314° 9.2		UNL / 8 000 (4 000) Class A, D, G				
△ BASEM 365037N 1275710E							
	134°/314° 12.5		UNL / 8 000 (5 000) Class A, D, G				
▲ BIGOB 364325N 1280952E							
	134°/314° 9.5		UNL / 8 000 (4 900) Class A, D, G				
△ YECHEON VOR(CUN) 363755N 1281931E							
	134°/314° 30.8	(34/34)	UNL / 8 000 (3 900) Class A, D, G				
△ ELAPI 362014N 1285051E							
	134°/314° 37.3		UNL / 8 000 (5 000) Class A, D, G				
△ POHANG VORTAC(KPO) 355838N 1292828E							
	107°/287° 17.2		UNL / 8 000 (3 500) Class A, D, G				
▲ BULGA 355609N 1294924E							
	107°/288° 44.4		UNL / 8 000 (1 500) Class A, D, G			↑	
▲ SAPRA(FIR BDRY) 354926N 1304325E							
INCHEON FIR							
FUKUOKA FIR							

Change : Information of track MAG for G585.

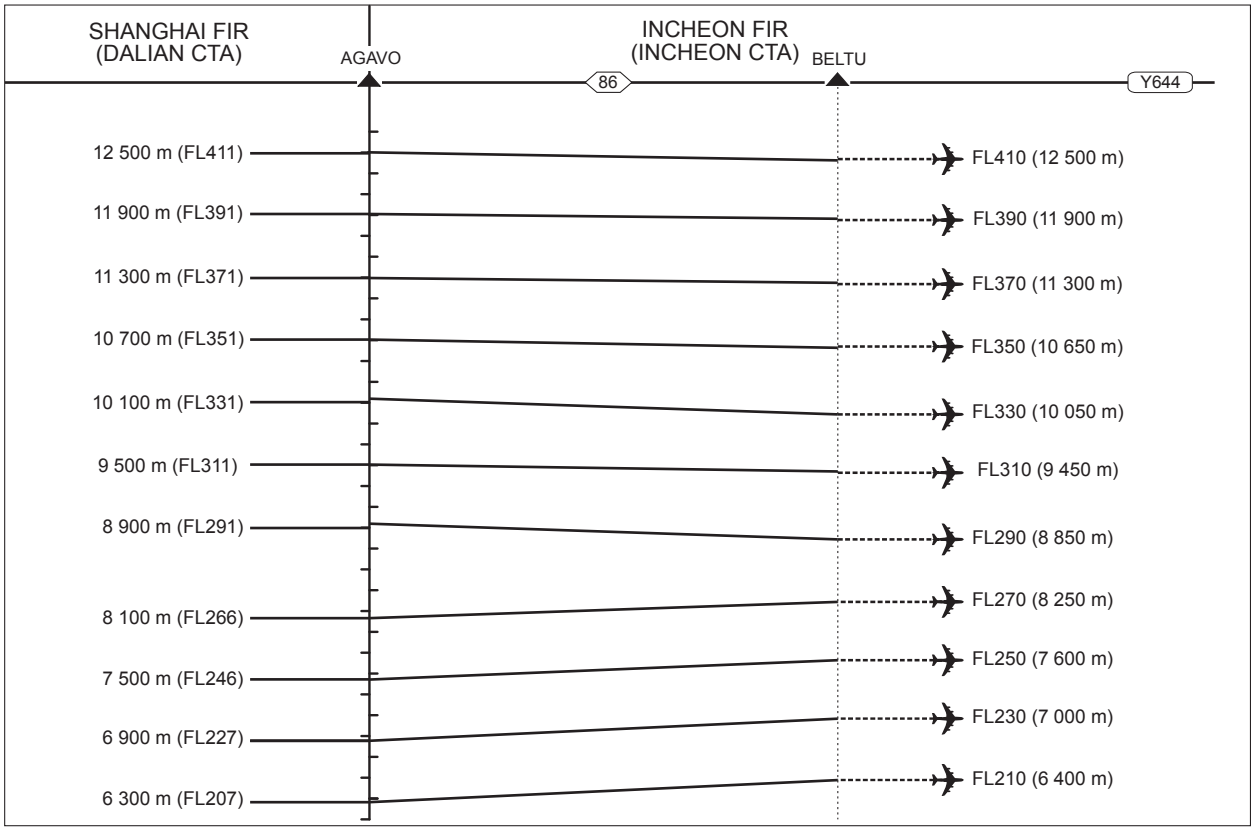
Route designator (RNP/RNAV) Name of significant points Coordinates	Track MAG ↑ / ↓ VOR RDL DIST (COP)		Upper limits / Lower limits or Minimum altitude(MOCA) ft AMSL or FL Airspace classification	Lateral limits NM	Direction of cruising levels Odd	Direction of cruising levels Even	Remarks Controlling unit channel Logon address
1	2		3	4	5		6
SHANGHAI FIR							
INCHEON FIR							
G597 ▲ AGAVO(FIR BDRY) 371000N 1240000E							**Daegu ACC**
	096° 276° 19.9		UNL FL 150 (1 500) Class A, D, G	10			FREQ : (at or below FL 295) 128.70(118.925) MHz 270.50(263.60) MHz
△ GONAV 371048N 1242453E	066° 246° 13.8						FREQ : (above FL 295) 132.80(120.525) MHz 290.60(335.45) MHz
▲ DANTI 371806N 1243929E	066° 247° 10.0						Westbound(SEL-AGAVO) FL 400, FL 380, FL 360, FL 340, FL 320, FL 300, FL 280, FL 260, FL 240, FL 220, FL 200.
△ ANSIM 372323N 1245009E	097° 278° 19.0						* Note G597 is only for Non-RNAV aircraft. Any aircraft approved for RNAV operations should use Y697.
▲ BINIL 372349N 1251359E	098° 278° 20.0						
▲ NOPIK 372412N 1253905E	098° 278° 41.0		UNL 8 000 (2 200) Class A, D, G				Only flying westbound from SEL to AGAVO on G597 is authorized. (Unless otherwise assigned by ATC, flying eastbound in this airway shall not be used.)
△ GOGET 372442N 1263036E	098° 279° 20.0		UNL 8 000 (3 400) Class A, D, G			↑	
△ ANYANG VORTAC(SEL) 372449N 1265542E	087° 267° 22.0		UNL 7 500 (3 500) Class A, D, G		↓		Aircraft flying eastbound from AGAVO to SEL on G597 shall get PPR 24 hours before from Incheon ACC.
△ EGOBA 372915N 1272246E	087° 268° 13.9		UNL 7 500 (5 100) Class A, D, G				REF. ENR 3.1-9 for the detailed altitude conversion procedures.
▲ KARBU 373159N 1273952E	088° 268° 22.9	(44/45)	UNL 7 500 (5 600) Class A, D, G				**Daegu ACC**
▲ TORUS 373625N 1280807E	088° 268° 21.8		UNL 7 500 (7 500) Class A, D, G				FREQ : (at or below FL 295) 134.175(123.65) MHz 272.40(233.60) MHz
▲ BIKSI 374032N 1283504E	088° 269° 8.2		UNL 7 500 (7 100) Class A, D, G				FREQ : (above FL 295) 122.250(125.925) MHz 263.350(263.85) MHz
△ GANGWON VORTAC(KAE) 374203N 1284514E	130° 310° 30.0		UNL 8 000 (7 100) Class A, D, G				Only flying westbound from LANAT to KAE on G597 shall get 24 HR PPR from Daegu ACC.
▲ PILIT 372631N 1291731E	130° 311° 31.2		UNL 9 000 (3 100) Class A, D, G				Airspace Classification refer to ENR 3.1-1
▲ ESNEG 371014N 1295051E	131° 311° 47.1		UNL 9 000 (1 500) Class A, D and G				
△ AGSUS 364521N 1304044E	131° 312° 42.9						
▲ LANAT(FIR BDRY) 362224N 1312542E						↑	
INCHEON FIR							
FUKUOKA FIR							

Change : Information of track MAG for G597.

ALTITUDE CONVERSION FROM INCHEON CTA TO DALIAN CTA (RVSM)



FROM DALIAN CTA TO INCHEON CTA



Change : Information of altitude conversion.

INTENTIONALLY

LEFT

BLANK

3.1.2 DOMESTIC ATS ROUTES

Route designator (RNP/RNAV) Name of significant points Coordinates	Track MAG ↑ / ↓ VOR RDL DIST (COP)	Upper limits / Lower limits or Minimum altitude(MOCA) ft AMSL or FL / Airspace classification	Lateral limits NM	Direction of cruising levels Odd	Direction of cruising levels Even	Remarks Controlling unit channel Logon address
1	2	3	4	5		6
V11						**Daegu ACC**
▲ PILIT 372631N 1291731E						
	183° 003° 19.3	UNL 9 000 (7 000) Class A, D, G	8		↓	FREQ : (at or below FL 295) 134.175(123.65) MHz 272.40(233.60) MHz FREQ : (above FL 295) 122.250(125.925) MHz 263.350(263.85) MHz 1. 11 000 ft to FL 240, at or above FL 280 will be blocked. 2. At or above 11 000 ft, required 15 days PPR from Air Traffic Management Office. 3. Airspace Classification refer to ENR 3.1-1
KAE R 128/30 DME KPO R 001/88 DME						
▲ NOBUT 370715N 1291957E	183° 003° 28.6	UNL 9 000 (6 400) Class A, D, G				
▲ OSVOM 363844N 1292331E	183° 003° 18.5	UNL 9 000 (4 000) Class A, D, G				**Daegu ACC** FREQ : 120.575(119.375, 119.325, 134.375) MHz 254.70(335.75) MHz 1. 11 000 ft to FL 240, at or above FL 280 will be blocked. 2. At or above 11 000 ft, required 15 days PPR from Air Traffic Management Office. 3. Airspace Classification refer to ENR 3.1-1
▲ LOSTO 362016N 1292548E	183° 003° 21.7	UNL 9 000 (3 300) Class A, D, G				
△ POHANG VORTAC(KPO) 355838N 1292828E	213° 033° 37.2	UNL 9 000 (4 200) Class A, D, G				
▲ APARU 352442N 1290932E	213° 033° 19.0	UNL 9 000 (4 400) Class A, D, G		↑		
△ BUSAN VORTAC(PSN) 350721N 1285958E						

Route designator (RNP/RNAV) Name of significant points Coordinates	Track MAG ↑ / ↓ VOR RDL DIST (COP)		Upper limits / Lower limits or Minimum altitude(MOCA) ft AMSL or FL Airspace classification	Lateral limits NM	Direction of cruising levels Odd	Direction of cruising levels Even	Remarks Controlling unit channel Logon address
1	2		3	4	5		6
V543							
△ DALSU 350731N 1264206E							
	098°/278° 5.4		UNL / 8 000 (2 500) Class A, D, G	10	↓		**Incheon ACC** FREQ : (at or below FL 255) 120.725(128.30) MHz 263.90(272.75) MHz FREQ : (above FL 255) 123.725(124.50) MHz 239.25(275.40) MHz Airspace Classification refer to ENR 3.1-1
△ GWANGJU VOR(KWA) 350734N 1264844E	098°/278° 2.6		UNL / 8 000 (2 000) Class A, D, G				
△ SAMUL 350736N 1265154E	098°/278° 22.1		UNL / 8 000 (5 200) Class A, D, G				
▲ TEDAN 350744N 1271852E	098°/278° 13.5		UNL / 8 000 (8 000) Class A, D, G				**Daegu ACC** FREQ : 128.175(128.325) MHz 335.50(275.20) MHz Between ANUBA and SAPDI 11 000 ft AMSL to FL 190 VMC-IMC use for training purpose. Airspace Classification refer to ENR 3.1-1
△ ANUBA 350746N 1273523E	098°/279° 44.7	(25/83)	UNL / 8 000 (8 000) Class A, D, G				
▲ SAPDI 350737N 1282952E	099°/279° 1.6		UNL / 8 000 (3 800) Class A, D, G				
▲ SARAM 350736N 1283147E	099°/279° 11.9		UNL / 8 000 (4 000) Class A, D, G				
△ ANKUS 350730N 1284616E	099°/279° 11.2		UNL / 8 000 (3 500) Class A, D, G			↑	
△ BUSAN VORTAC(PSN) 350721N 1285958E							
V547							**Incheon ACC**
△ GWANGJU VOR(KWA) 350734N 1264844E	073°/253° 54.7	(48/48)	10 000 / 9 000 (8 600) Class D	10	↓		FREQ : (at or below FL 255) 120.725(128.30) MHz 263.90(272.75) MHz FREQ : (above FL 255) 123.725(124.50) MHz 239.25(275.40) MHz Airspace Classification refer to ENR 3.1-1
▲ IGDOK 353104N 1274907E	073°/254° 41.6		10 000 / 9 000 (5 000) Class D			↑	**Daegu ACC** FREQ : 128.175(128.325) MHz 335.50(275.20) MHz Airspace Classification refer to ENR 3.1-1
△ DALSEONG VORTAC(TGU) 354835N 1283527E							

Change : Information of remarks for V543.

Route designator (RNP/RNAV) Name of significant points Coordinates	Track MAG ↑ / ↓ VOR RDL DIST (COP)		Upper limits / Lower limits or Minimum altitude(MOCA) ft AMSL or FL Airspace classification	Lateral limits NM	Direction of cruising levels Odd	Direction of cruising levels Even	Remarks Controlling unit channel Logon address
1	2		3	4	5		6
V549							
△ GUNSAN VORTAC(KUZ) 355437N 1263641E							**Incheon ACC**
	101°/282°			10	↓		
△ ELPOS 355410N 1264707E	8.5		10 000 / 7 000 (1 800) Class D				FREQ : (at or below FL 255) 126.175(134.375) MHz 317.85(335.55) MHz
	102°/282°						FREQ : (above FL 255) 132.15(123.55) MHz 263.15(272.60) MHz
△ RINBO 355352N 1265349E	5.5						
	102°/282°	(49/48)	10 000 / 7 000 (3 500) Class D				Airspace Classification refer to ENR 3.1-1
△ MELES 355251N 1271542E	17.8						
	102°/282°		10 000 / 7 000 (5 000) Class D				
△ OPEDA 355149N 1273652E	17.2						
	102°/283°		10 000 / 7 000 (6 600) Class D				**Daegu ACC** FREQ : 125.375(125.775, 124.575) MHz 234.15(317.35, 335.50) MHz
△ DALSEONG VORTAC(TGU) 354835N 1283527E	47.7						
	085°/266°		UNL / 6 000 (5 400) Class A, D, G				Airspace Classification refer to ENR 3.1-1
▲ LAPAL 355413N 1290452E	24.6						
	085°/266°						TGU is used between TGU and KPO.
△ POHANG VORTAC(KPO) 355838N 1292828E	19.7					↑	
W45							
△ GWANGJU TACAN(KWJ) 350723N 1264810E							**Incheon ACC**
	098°/278°			10	↓		
△ RIMPO 350739N 1273502E	38.4		UNL / 8 000 (8 000) Class A, D, G				FREQ : (at or below FL 255) 120.725(128.30) MHz 263.90(272.75) MHz FREQ : (above FL 255) 123.725(124.50) MHz 239.25(275.40) MHz
	098°/279°	(23/85)	UNL / 8 000 (8 000) Class A, D, G				Airspace Classification refer to ENR 3.1-1
▲ RUNIT 350734N 1282952E	45.0						**Daegu ACC**
	099°/279°		UNL / 8 000 (4 000) Class A, D, G				FREQ : 128.175(128.325) MHz 335.50(275.20) MHz
△ BUSAN VORTAC(PSN) 350721N 1285958E	24.7					↑	Between RIMPO and RUNIT 11 000 ft AMSL to FL 190 VMC-IMC use for training purpose. Airspace Classification refer to ENR 3.1-1

Change : Information of remarks for W45.

Route designator (RNP/RNAV) Name of significant points Coordinates	Track MAG ↑ / ↓ VOR RDL DIST (COP)	Upper limits / Lower limits or Minimum altitude(MOCA) ft AMSL or FL / Airspace classification	Lateral limits NM	Direction of cruising levels Odd	Direction of cruising levels Even	Remarks Controlling unit channel Logon address
1	2	3	4	5		6
W61						
△ SONGTAN VORTAC(SOT) 370540N 1270154E						**Daegu ACC**
	316° / 136°	UNL / 8 000 (2 400)	10		↓	FREQ : (at or below FL 295) 128.70(118.925) MHz 270.50(263.60) MHz
▲ MONSI 371247N 1265015E	11.7	Class A, D, G				FREQ : (above FL 295) 132.80(120.525) MHz 290.60(335.45) MHz
	316° / 136°	UNL / 8 000 (3 200)				
△ GOGET 372442N 1263036E	19.7	Class A, D, G		↑		Airspace Classification refer to ENR 3.1-1
W62						
△ SONGTAN VORTAC(SOT) 370540N 1270154E						**Daegu ACC**
	044° / 224°	UNL / FL 140 (4 600)	10	↓		FREQ : (at or below FL 295) 128.70(118.925) MHz 270.50(263.60) MHz
△ EGOBA 372915N 1272246E	28.8	Class A, D, G			↑	FREQ : (above FL 295) 132.80(120.525) MHz 290.60(335.45) MHz
						Airspace Classification refer to ENR 3.1-1
W526						
△ DALSEONG VORTAC(TGU) 354835N 1283527E						
	193° / 013°	UNL / 5 000 (4 900)	10		↓	**Daegu ACC**
▲ MASTA 352847N 1283340E	19.8	Class A, D, G				FREQ : 125.375(125.775, 124.575) MHz 234.15(317.35, 335.50) MHz
	193° / 013°	UNL / 5 000 (3 800)				
▲ SARAM 350736N 1283147E	21.2	Class A, D, G				Airspace Classification refer to ENR 3.1-1
	193° / 013°	UNL / 5 000 (3 200)				
▲ TOPAX 344555N 1282952E	21.7	Class A, D, G		↑		